

Telecom Customer Churn: Strategic Analysis & Retention Roadmap

Executive Summary

This analysis identifies the primary drivers of customer attrition within the telecommunications base. By evaluating contract structures, service tiers, and payment behaviours, high-risk customer segments were isolated. The most significant churn occurs very early in the customer lifecycle, specifically among month-to-month subscribers and Fiber Optic users. Shifting the organizational focus toward long-term contract incentivization and early-stage onboarding can significantly improve overall retention and customer lifetime value (LTV).

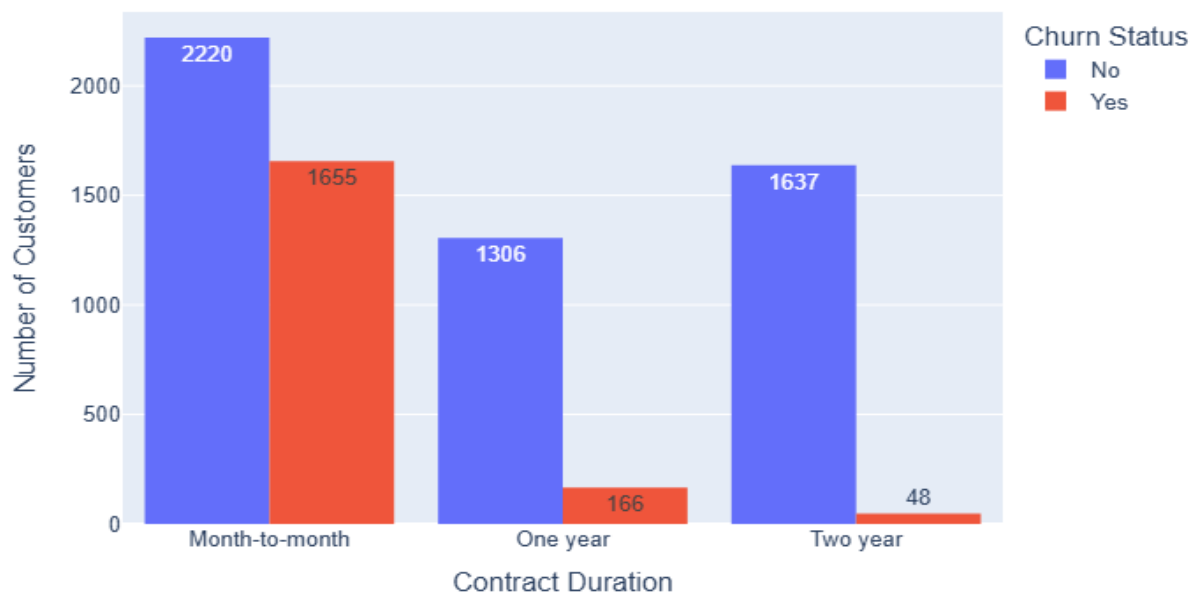
Data Preparation & Methodology

To ensure strict data integrity prior to visual analysis, the raw dataset underwent rigorous cleaning. Missing or blank values in financial columns (specifically TotalCharges) were handled, categorical variables were standardized for accurate grouping, and continuous variables like tenure were evaluated for outliers. This foundational processing ensures that all subsequent business metrics are derived from a highly accurate baseline.

Key Analytical Insights

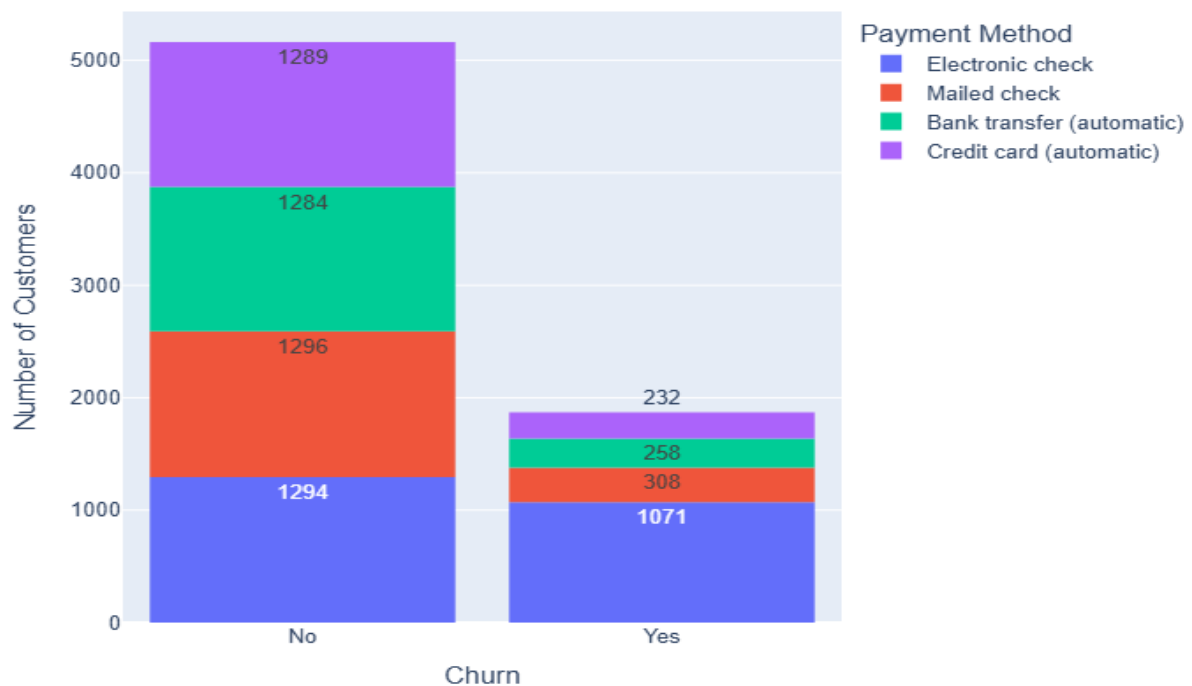
- **Contract Vulnerability:** Month-to-month contracts exhibit an exceptionally high churn volume compared to 1-year and 2-year commitments. These customers operate with zero friction to leave and lack long-term retention incentives.
- **The Premium Service Paradox:** Fiber Optic internet customers churn at a disproportionately higher rate than DSL users, despite Fiber being a premium, higher-speed tier.
- **Payment Method Friction:** Users paying via "Electronic Check" have the highest churn rate by a wide margin compared to automated methods (Credit Card or Bank Transfer).
- **Early-Stage Attrition:** Density distributions of tenure and total charges reveal a massive churn spike in the first few months. If a customer survives the initial 6-month window, their probability of long-term retention increases exponentially.

Customer Churn by Contract Type

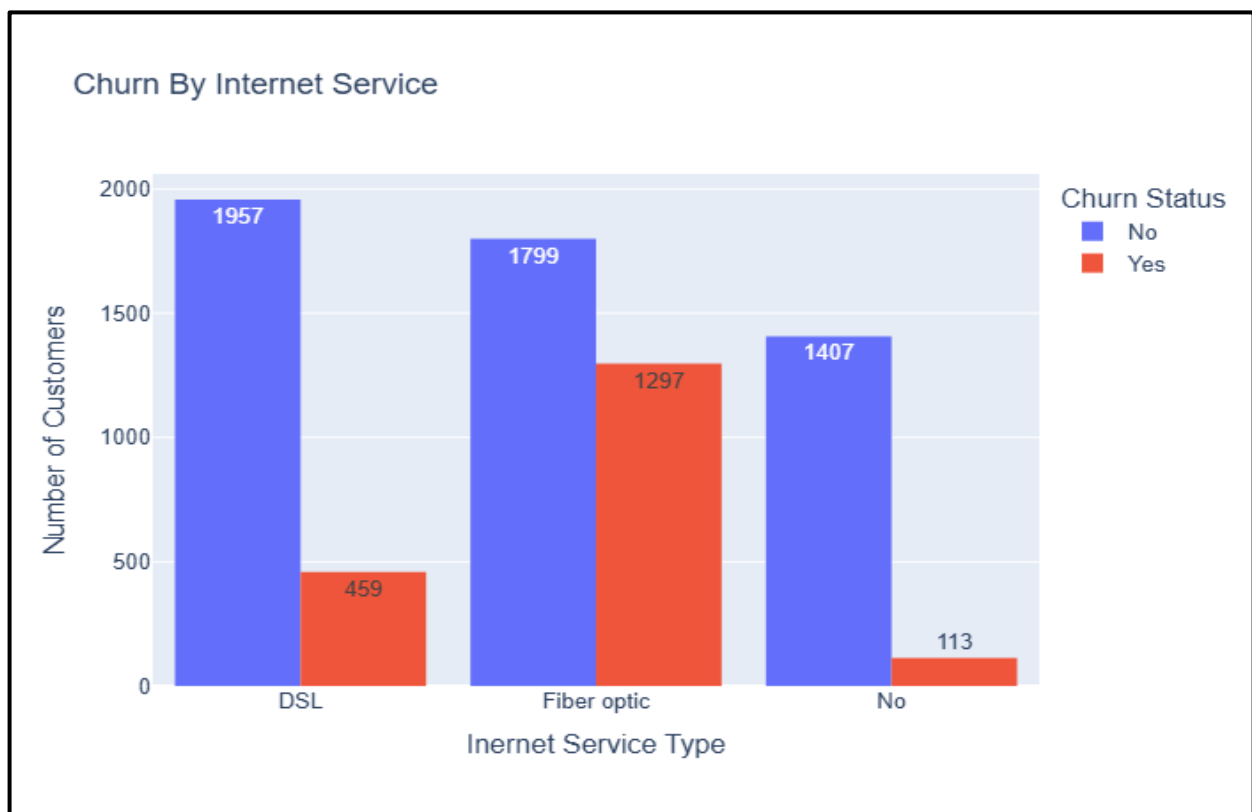


- Month-to-month contracts drive the overwhelming majority of churn (1655 customers, ~43% churn rate), while one-year and two-year contracts show dramatically lower churn (~11% and ~3%), highlighting contract lock-in as a strong retention lever.

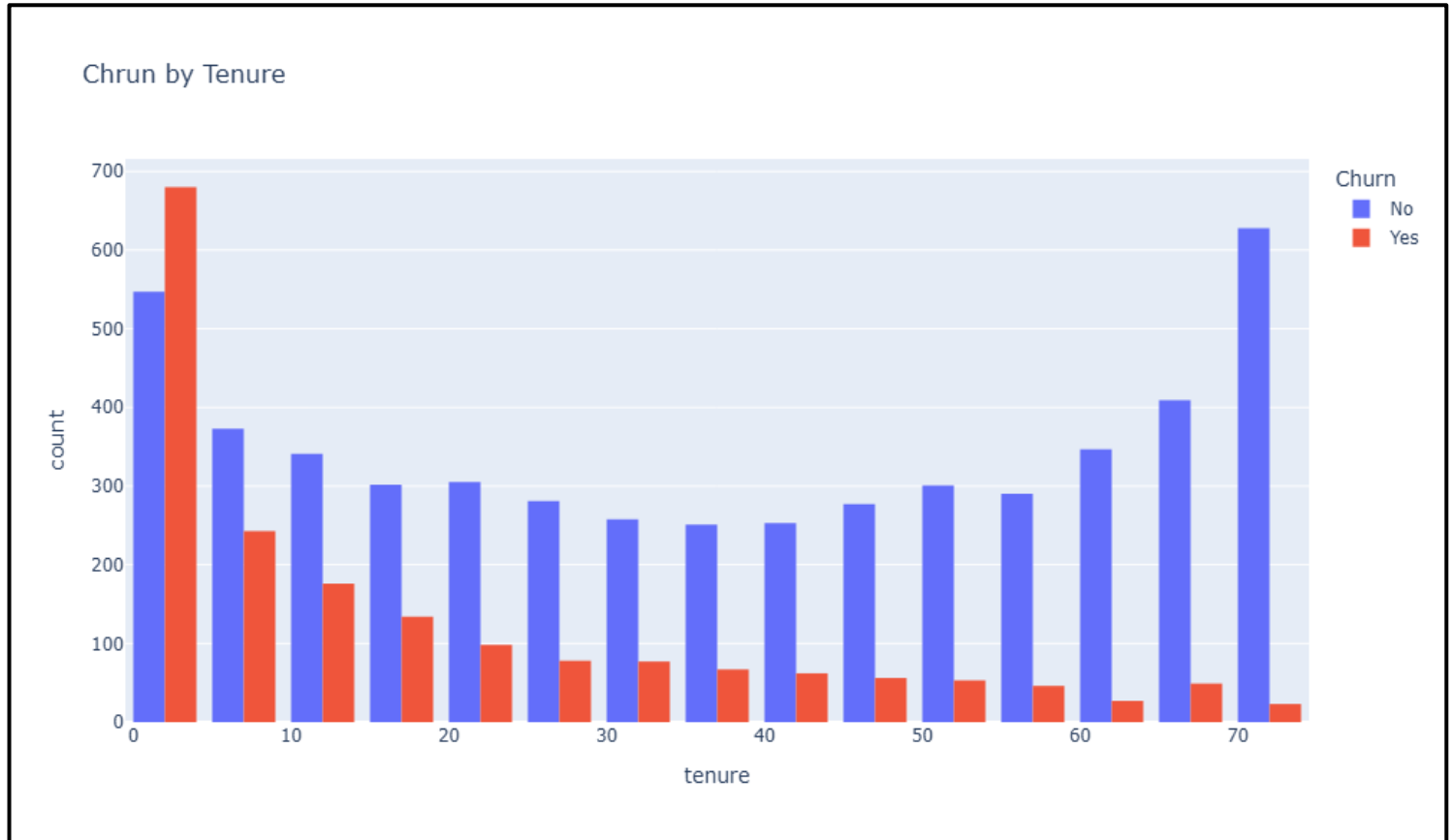
Customer Churn by Payment Method



- Electronic check users account for the largest share of churners (1071 out of 1861), churning at a disproportionately higher rate than automatic payment methods (bank transfer/credit card), indicating manual payment friction correlates with attrition.



- Fiber optic customers churn at a much higher rate (~42%) compared to DSL (~19%) and customers with no internet service (~7%), suggesting fiber service quality, pricing, or competition issues.



- Churn is heavily concentrated among new customers (0-10 months), with churners even outnumbering non-churners in the first bucket, while long-tenured customers (60+ months) show strong loyalty with minimal churn.

Actionable Business Recommendations

Risk Area	Strategic Action	Expected Business Impact
High Month-to-Month Churn	Deploy targeted campaigns offering immediate discounts or free service upgrades (e.g., free premium streaming add-on) for transitioning to a 1-year contract.	Stabilizes predictable monthly recurring revenue (MRR) and directly targets the largest exiting cohort.
Fiber Optic Attrition	Conduct a technical audit of the Fiber network in high-churn regions and trigger automated CSAT surveys within 14 days of Fiber installation.	Determines whether the premium churn is driven by technical unreliability, poor installation experiences, or competitor pricing.
Early Lifecycle Drop-off	Design a proactive 90-day onboarding sequence. Offer a one-time bill credit if the user enables Auto-Pay via Credit Card during month one.	Migrates users away from high-churn manual payment methods (Electronic Checks) and builds loyalty during the most vulnerable tenure phase.
Manual Payment Friction (Electronic Checks)	Incentivize the adoption of automated payments. Offer a recurring \$5 monthly bill discount or a one-time account credit for users who switch from Electronic Checks to Credit Card/Bank Transfer Auto-Pay.	Minimizes involuntary churn from forgotten payments and increases customer "stickiness" by reducing monthly billing friction

➤ Other insights from the data

- 26.5% of the Customers Churned.
- Customers without Online Security and Tech Support have higher churn rate.
- Customers without Device Protection and Online Backup have high churn rate.
- Customers With higher monthly charges are more likely to churn.
- Customers without Partner or Dependent have comparatively higher churn rate.

